HACKATHON 2026

Asset management

Final Challenge Instructions

McGill-FIAM Asset Management
Hackathon Instructions

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1. Introductory Note

Congratulations on reaching the finals. Out of a highly competitive field you are one of eleven finalist teams — and the number itself is worth a word of explanation. The finals have been capped at ten in previous editions. This year the quality of the first-round submissions was high enough that we expanded the field, which is the clearest statement we can make about the standard of work this cohort produced.

The first challenge asked you to build a market-neutral U.S. equity strategy on a monthly panel, using 147 firm characteristics and a corpus of 8-K filings. Many of you did that well. The finals ask a harder question, and a more realistic one: what happens to your strategy when the decision frequency goes from monthly to daily, when the news arrives as a stream rather than a file, when the universe you may short is constrained the way a real prime-brokerage relationship constrains it, and when your results must be reported the way an allocator expects to receive them?

Three things change materially. You may now trade daily, and we provide daily returns for every stock in the universe. You gain access to the Carbon Arc data platform, where all finalists will be onboarded, and with it a large library of alternative and market data. And the short leg now has a hard floor on market capitalisation, because the micro-cap short books that several first-round submissions produced are, in practice, unborrowable.

Everything else about the mandate is unchanged. The universe is U.S. public equities. The strategy is equity market neutral. The benchmark is cash plus a premium, and the headline number is the information and Sharpe ratios. You are building the same product; you are now building it under conditions closer to the ones it would actually face.

This document contains what you need to start work. Further constraints and the final data configuration will be given at the start of the two-day event. Read the forward-looking bias and reporting sections carefully: at daily frequency the ways to leak future information multiply, and we will be checking.

DATA ACCESS

Online folder: **MCGILL FIAM ASSET MANAGEMENT HACKATHON FINALS OCTOBER 2026**

Carbon Arc: every team member is onboarded individually.

Arcana: one account per team, registered to the team leader's email address.

Watch your inbox. Onboarding emails from Carbon Arc and Arcana will arrive over the coming days, Carbon Arc first and Arcana at the same time or shortly after. Check your spam folder, and act on them when they arrive. A team that has not completed onboarding before the event starts loses working hours it cannot get back.

Full contents of the folder are listed in Section 11.

2. What Changes from the First Challenge

The table below is the complete list of differences. If a row is not in this table, assume the first-challenge rule still applies.

	First Challenge	Final Challenge
Universe	U.S. public equities	Unchanged — U.S. public equities only
Return data	Monthly returns	Daily returns for all stocks, in addition to monthly
Rebalancing	Monthly or slower; at least semi-annual	Any frequency from daily to monthly
Shortable universe	No restriction	Market capitalisation ≥ US$500 million at formation
Long universe	No restriction	Unchanged — no restriction
News and text	8-K filings corpus	8-K corpus plus the Carbon Arc data library
Platform	Team's own environment	Carbon Arc and Arcana; teams onboarded on both. Team's own coding environment
Fund size	Not specified	US$1 billion initial endowment, compounding over the test period
Position count	100–500 names, long and short combined	200–500 names, long and short combined
Single position limit	Maximum 10% of capital	Maximum 5% of capital
Agentic work	Encouraged, mainly in signal generation	Extended to portfolio construction and position-level risk management (Section 5)
Mandate	Market neutral, gross ≤ 200%, net within ±50%	Unchanged
Benchmark	3-month T-bill + 4% per year	Unchanged
Headline metric	Sharpe, Information ratios	Unchanged, with standardised reporting (Section 9)
Reporting	Prescribed statistics	Standardised industry-format table, turnover mandatory
Evaluation	Code review for look-ahead bias	Economic and factor review on the Arcana platform (Section 11)

A note on what is not changing. The point of the finals is not to reward whoever adds the most new data. A clean, well-understood monthly strategy that you extend thoughtfully to daily/weekly decisions and report honestly will beat a strategy that bolts on six alternative datasets nobody had time to validate. Several first-round submissions were undone not by weak ideas but by data handling. Carbon Arc is an opportunity, not an obligation.

3. The Final Mandate

Design and test an equity market-neutral U.S. equity strategy that uses machine learning, large language models, agentic workflows, or any combination of them, to forecast returns or firms' fundamentals and to construct a portfolio from those forecasts. Report performance strictly out of sample.

3.1 Portfolio constraints

Constraint	Specification
Starting capital	US$1 billion endowed at the start of the out-of-sample period; the fund compounds from there
Positions	At least 200 and at most 500 names, long and short legs combined
Gross exposure	Up to 200% of capital (leverage of at most 2×)
Net exposure	Within −50% to +50% at all times; dollar neutrality is the natural centre
Shortable universe	Only names with market capitalisation ≥ US$500m, measured at the formation date
Single position	Maximum 5% of capital in any one name
Rebalancing	Any frequency from daily to monthly; you choose and you justify it
Benchmark	3-month U.S. Treasury bill + 4% per year, computed month by month
Out-of-sample window	01/2021 – 08/2026

3.2 A note on the position count

The floor has doubled, from 100 names to 200. The ceiling is unchanged at 500. The effect is to narrow the range rather than widen it, and the narrowing is deliberate.

The higher floor is about breadth. The fundamental law of active management says that your information ratio scales with the square root of the number of independent bets you take, for a given level of skill per bet. A market-neutral book is a diversified product by construction, and a 100-name book run at 200% gross leaves individual positions large enough that a handful of names can dominate the result — which is a concentrated stock-picking fund, not a systematic strategy. Two hundred names is the point at which the statistics of your signal, rather than the idiosyncrasies of a few holdings, start to drive the outcome.

The ceiling stays where it was because 500 names is already enough to express almost any signal you are likely to build on this universe, and because every name you add is a name you have to trade, monitor and pay for. At 500 names and 200% gross, equal weighting gives positions of roughly 0.4% of capital each — about US$4 million at the stated fund size; at the 200-name floor, roughly 1%, or US$10 million. Against that, the single-position cap has been tightened from 10% to 5% of capital, which is US$50 million. At these position counts the

cap is a backstop rather than a binding constraint, and a book that comes anywhere near it is concentrated in a way this mandate is not meant to accommodate — expect to be asked to justify it.

Two interactions to keep in view. First, breadth and turnover trade against each other: a 500-name book rebalanced daily is a large number of trades, and at US$1bn each of those trades is large enough in its own right that market impact, not commission, is the binding cost. Second, the short leg now draws on a restricted universe. Roughly 3,000 to 3,500 U.S. listed companies carry a market capitalisation above US$500m, so a 250-name short book is shorting something like seven or eight per cent of everything you are permitted to short. That is a workable fraction, but it means your short-side alpha has to come from selection within a pool you are already sampling heavily.

3.3 Fund size, capacity and market impact

This year the mandate carries a stated fund size. You are endowed with US$1 billion at the start of the out-of-sample period, and the job is to grow it. The fund compounds: whatever the portfolio earns or loses in one period is capital you are managing in the next, exactly as it would be for a real fund. There are no subscriptions and no redemptions, so the only thing that changes your capital is your own performance.

The first challenge left fund size open, and that omission let several teams report returns from books that could not have been traded at any institutional scale. Naming the number closes that gap, makes every submission comparable on the same terms, and gives you a second figure that an allocator actually cares about: what the billion dollars was worth at the end.

Work through what it implies at inception. At 200% gross you are deploying US$2 billion — roughly US$1 billion long and US$1 billion short. Spread across 500 names that is about US$4 million a position; across 200 names, about US$10 million. Those are not small orders. A US$4 million position in a company that trades US$20 million a day is a fifth of a day's volume, and professional desks generally keep participation below ten to twenty per cent of average daily volume to avoid moving the price against themselves. Several of the names that looked attractive in your first-round backtests will not absorb this.

And because the fund compounds, this problem does not stay still. Gross and net exposure are percentages of current capital, so every position scales with the fund. A strategy that doubles the endowment has also doubled every order it sends, and names that comfortably absorbed US$4 million at inception may not absorb US$8 million five years later. Success tightens your own capacity constraint. This is not a trick of the exercise — it is the central operational problem of a successful fund, and a team that notices it in the backtest and says what it would do about it is demonstrating something most candidates never think about.

This is also the real justification for the US$500 million market-capitalisation floor on the short side. It is not an arbitrary restriction invented to make the problem harder. At a billion dollars of capital, a short book built from micro-caps is not merely expensive to borrow — it is unborrowable in the size the mandate requires, and a backtest of it is fiction. The floor forces the short leg into names where the position can actually exist.

Neutrality has more than one definition and you should be explicit about which one you enforce — dollar, beta, or sector and factor neutrality. A dollar-neutral book that is long high-beta names and short low-beta names is not neutral; it is a levered long position in disguise. We will estimate your realised beta ourselves, at both daily and monthly frequency, and a beta materially different from zero over the out-of-sample period will be read as a directional strategy however the deck describes it.

3.4 Why the benchmark is cash plus a premium

A market-neutral book is nominally zero-cost: short proceeds finance the longs, and the capital posted as

collateral earns the bill rate. Benchmarking it against the S&P 500 makes no sense, because a manager who matched the index would have failed at the mandate. Your hurdle is the return you could have had risk-free plus the premium an allocator pays for active management. With the 3-month bill near 3.7% the hurdle is roughly 7.7% annualised; in 2021, when bills yielded almost nothing, the same rule gave a hurdle near 4%. It is a time series, not a constant.

Show the S&P 500 alongside your strategy for context and for the beta regression, but the number that decides whether the strategy worked is the information ratio of your return against cash plus 4%.

4. Daily Decisions

We provide daily returns for every stock in the universe, and you may rebalance as often as every trading day. This is a genuine extension of the problem rather than a convenience, and it is worth being clear about what it buys you and what it costs.

What it buys you

- Event response. A material disclosure or a news event is a dated, discrete thing. At monthly frequency you can only trade it at the next month-end, by which point most of the price response has happened. At daily frequency you can act on it in the window where it is still information.
- Faster-decaying signals. Several of the characteristics in the panel, and most of what alternative data captures, have half-lives measured in days or weeks. Monthly rebalancing throws that away.
- Tighter risk control. Beta, sector and factor exposures can be brought back to target daily rather than drifting for a month.

What it costs

- Turnover. This is the binding constraint and it is not close. A strategy that replaces a tenth of its book every day has an annualised one-way turnover above 2,500%, and an average holding period of under two weeks. At institutional size that is not a strategy, it is a transfer of your alpha to the market makers. Report turnover honestly (Section 9) and expect the judges to ask about it.
- Cost sensitivity. At monthly frequency, commissions and spreads are a rounding error against a good signal. At daily frequency they are often the whole answer. If you trade daily you must model costs, and you must report performance before and after them.
- Noise. Daily cross-sectional return dispersion is dominated by microstructure and liquidity, not by fundamentals. A model trained to predict tomorrow's return is largely being trained to predict noise. Think about whether your signal has a daily horizon at all before you trade it daily.
- Point-in-time discipline. Every leakage hazard in Section 7 gets sharper when the decision interval shrinks from twenty-one trading days to one.

You are not required to trade daily. Daily trading is permitted, not mandated. A defensible answer is a weekly or monthly core book with a daily overlay for events — or a monthly book that you simply execute better. "We tested daily rebalancing, it did not survive costs, and here is the evidence" is a strong finding and we will credit it. What we will not credit is a daily strategy whose reported turnover and costs are missing or implausible.

5. Agentic Workflows and Position-Level Risk Management

Almost every first-round submission that used agents used them in the same place: on the signal. An agent read a filing, classified an event, scored a company, and the output became a factor. That work was often good, and you should continue it. But it is now the obvious move, which means it is no longer where the differentiation lies.

Two less-explored opportunities are where we want to see effort this time. The first is downstream of the signal: portfolio construction and risk management. The second is upstream of it: using agents to explore an unfamiliar data library and find signals nobody is running yet. Both are a long way from "have a model read a filing and score it", and both are where this year's differentiation will come from.

Take the risk layer first. This is deliberate. Signal generation is a well-defined supervised learning problem with a large literature and mature tooling. Position sizing and risk control are neither. It is the part of a systematic strategy that practitioners spend most of their time on, where most strategies actually fail, and where the governing logic is a tangle of soft constraints, conditional exceptions and trade-offs that no single objective function captures cleanly. That is precisely the shape of problem a reasoning system can be good at — and it is almost entirely unexplored in student competitions.

5.1 What an agent can usefully do in the risk layer

These are concrete, buildable, and each one is testable against a fixed-rule alternative. You do not need all of them. One done well, with evidence, is worth more than six gestured at.

1. **Pre-trade constraint review.** Before each rebalance, hand the agent the proposed book, the current book, and the full constraint set — gross, net, position count, the 5% cap, the US$500m short floor, sector and factor exposures, realised beta. Its job is not to approve or reject but to explain: which constraint binds, by how much, what the cheapest remedy is, and what that remedy costs in expected return. A rule-based optimiser returns a number. This returns a reason, and the reason is what you put in the deck.
2. **Exposure attribution and drift detection.** Decompose the book each day into intended and unintended exposures. An intended exposure is one somebody decided to take; everything else is drift. A short book that quietly migrates into small-cap value over six months, or a long book whose sector weights track the signal's own coverage bias, is carrying risk nobody chose. An agent that surfaces this in plain language, with the attribution behind it, is doing work a risk analyst does.
3. **Context-conditional position sizing.** A fixed sizing rule treats every one-sigma signal identically. It does not know that this name reports earnings in three days, that one is hard to borrow, a third has had six news events in a week, and a fourth has seen its realised volatility triple. An agent can condition the size on that context — smaller into a print you have no edge on, smaller where borrow is tight, smaller where the position would take too many days to exit, smaller where the signal and the crowd agree too enthusiastically. This is, in our view, the single highest-value application available to you.
4. **Trade triage against the cost model.** Most rebalancing trades are noise: a name moved from rank 103 to rank 97 and the optimiser dutifully adjusts the weight. Give an agent the target book, the current book and the cost model, and have it decide which trades are worth paying for and which should be left alone. At this fund size this is not a refinement — it is where a daily strategy either survives its own turnover or does not.
5. **Anomaly escalation.** When a position loses materially more than the model expected, something has usually changed — in the company, in the data, or in your pipeline. An agent that investigates the exception, distinguishes a genuine company event from a stale feed or a corporate action you mishandled, and escalates with a recommendation, is catching the class of error that quietly destroys live strategies.
6. **The post-trade explainer.** A short written account, each period, of what moved the book and why: which signals drove the turnover, which exposures changed, which positions contributed. Allocators ask for exactly this, and very few students produce it. It is also the most honest test of whether you understand your own

strategy.

5.2 Agents on the data itself: exploratory signal discovery

The second under-explored frontier is upstream of the signal, not downstream of it. The Carbon Arc library runs to seventy-seven datasets. Reading the documentation, forming hypotheses, pulling samples, checking coverage and testing candidate features across that library is several weeks of analyst work.

This is where the platform's delivery mechanics matter more than they first appear. Carbon Arc exposes a REST API **and an MCP server**, which means a language model can query the data directly rather than waiting to be handed a feature matrix someone built by hand. The integration is close to plug-and-play: you are not writing connectors, you are pointing a model at a data source and asking it questions. What that buys you is the ability to explore far more of the library than you could manually in the time available. **NOTE**, once onboarded, the platform has 20K tokens per day for the whole competition starting from the release. It is normally enough for the unlimited download but if you hit the limit, expect the system to reset the next day, and renew upload next day.

Four applications worth building, roughly in order of expected payoff:

1. **Automated dataset triage.** Point an agent at the catalogue and at your universe, and have it report back per asset: what fraction of your names it covers, when the history starts, what the publication lag is for the metric you care about, what the entity key is, and a first-pass judgement on whether a usable feature exists. This is the four-question screen from Section 6.6, run across seventy-seven assets instead of three. It produces a written record you can put straight into the deck.
2. **Hypothesis generation at scale.** Give an agent a dataset, a sector and the question: what in here would plausibly predict returns, and why? Have it propose candidate features together with an economic rationale for each. The agent generates; your walk-forward framework adjudicates, on validation data only. The requirement for a stated economic rationale is what separates this from data mining — make it mandatory, and reject candidates that cannot supply one.
3. **Cross-dataset combinations.** The commercial value of these datasets is mostly realised one at a time. The genuinely under-explored space is the joins: job postings against software adoption, foot traffic against card spend in the same catchment, open-source development activity against engineering hiring, government contract awards against facility-level capacity. An agent can enumerate and screen candidate pairs far faster than you can, and these combinations are where a signal nobody else is running is most likely to be hiding.
4. **Documentation archaeology.** Every asset ships a field-level data dictionary with types, descriptions and sample values. Models are good at reading these closely and surfacing the field that actually matters — the exclusivity flag on the news table, the claim-maturity column, the record-version key — which is exactly what a human skimming a schema at speed will miss. Several of the points in Section 6 were found this way.

Why this is a real edge, and where it stops being one

Be clear-eyed about why the opportunity exists. Datasets like these are licensed mostly to institutional users who apply them to a handful of names in a sector they already cover. Systematic, cross-sectional use across an entire universe is far less common, historically because the entity mapping was too expensive for anyone but the largest shops to build. Carbon Arc has done the mapping, and agents make the exploration cheap. That combination is new enough that the obvious signals in this library are genuinely not all arbitraged away — which is not something one can say about the 147 characteristics in the main panel.

The warning that comes with it. Cheap exploration means cheap multiple testing, and this is the way an agentic research pipeline fails. If an agent proposes and tests four hundred candidate features and you report the best one, you have found noise — and you have found it with great confidence, because the search was thorough. Three defences, and we expect all three: state how many hypotheses you tested, in total, not how many you kept; hold out a slice of data that nothing touched until the very end; and require an economic rationale written down before the test, not after. A team that reports "we tested 380 candidates, 11 survived validation, 4 survived the holdout, here they are" is doing science. A team that shows four features and no denominator is not, and we will ask for the denominator.

5.3 Design requirements

These apply to everything in this section — the risk layer and the data-exploration work alike. If you build an agent into the pipeline, they are not suggestions. A risk layer that cannot be audited is not a risk layer.

1. **The agent proposes; code enforces.** Every hard constraint — gross, net, position count, the 5% cap, the US$500m short floor — must be enforced in deterministic code after the agent has had its say. Never in a prompt. A language model that is asked nicely to respect a limit will eventually not, and a breach you discover in the backtest is a breach the judges will discover too.
2. **Log every decision with its inputs.** What the agent saw, what it decided, and what it gave as a reason. This log is the deliverable as much as the returns are, and it is what lets a reviewer confirm that the agent was not reasoning from information it should not have had.
3. **Point-in-time applies to agents too.** Everything in Section 8.2 — dated inputs only, no live retrieval, declared model cutoff — applies with full force here. An agent sizing a position on the strength of knowledge about how that company's quarter turned out has leaked, and no amount of care elsewhere repairs it.
4. **Build the counterfactual, and report it.** Run the identical strategy twice: once with the agent in the risk layer, once with a fixed rule in its place. Report both. Without that comparison you have no basis for claiming the agent contributed anything, and we will not take the claim on faith. This is the single most important requirement in this section.
5. **Design the trigger, not just the agent.** A model call per name per day across 500 names and 1,400 trading days is roughly 700,000 calls. That is neither affordable nor reproducible. Build the agent to run on exceptions — a constraint near its bound, an unexplained loss, a sudden change in a name's context — and handle the routine cases in code. Knowing when not to invoke the model is itself a design skill.
6. **Cache and version every response.** Model outputs are non-deterministic. Persist each response keyed by its inputs so the pipeline reruns identically, and so a judge can read exactly what the model said on any given date.

A negative result here is a real result. "We built an agentic risk layer, measured it against a fixed rule, and it did not improve the information ratio — here is where it helped, here is where it hurt, and here is our read on why" is a strong, publishable finding and the judges will credit it accordingly. What earns nothing is an agent in the architecture diagram with no evidence that it changed anything.

6. The Carbon Arc Data Platform

All finalists will be onboarded on Carbon Arc, our data partner for the finals — www.carbonarc.co, with the catalogue at www.carbonarc.co/explore-data.

Carbon Arc aggregates a large library of alternative and market datasets and — this is the part that matters for your work — resolves them against a single entity ontology, so that a merchant, a brand, a physical store, a software product or a code repository can be mapped to a listed company without you building a separate crosswalk for every vendor.

Data reaches you four ways: bulk extracts, pre-aggregated views, a REST API, and an MCP server that an LLM agent can query directly. That last route is worth your attention given this year's theme: an agent can be given a question about a company and go and interrogate the data itself rather than being handed a pre-built feature matrix. Section 5.2 sets out how to use that productively.

Alongside this document you will receive **MCGILL_2026_CarbonArc_Data_Screen.xlsx**, a screening workbook covering the full catalogue with, for each asset, its identifier, provider, entity key, history, update frequency, publication lag, region, a usability verdict for this mandate, and an assessment of survivorship exposure. Treat it as the manual. It will save you most of a day.

6.1 What Carbon Arc is good at

- **Entity resolution.** The hardest part of alternative data is joining it to a security. Carbon Arc has done that work. You still have to verify it (Section 6.3), but you are verifying a mapping rather than building one.
- **Breadth across the real economy.** Card spending, point-of-sale, foot traffic, clickstream, job postings and job movements, software adoption, open-source development activity, medical and pharmacy claims, government contracts, trade flows, facility-level physical risk. These are reads on company performance that do not come from the accounting statements, which is exactly why they can carry information prices do not.
- **Documented lag, and in places documented maturity.** Each asset publishes its update frequency and publication lag. A few go further: the medical claims asset publishes its completion curve — roughly 45% of a service day's claims arrive in week one and 80% by week four — which makes a correct point-in-time implementation possible rather than guesswork. Prefer assets that tell you this.
- **Record versioning.** Many assets carry last-updated, hash-key and partition-date columns. These exist because records get revised. Used properly they let you reconstruct what the data looked like on a past date, which is the single most valuable thing an alternative data platform can give a backtester.
- **A handful of genuinely long histories.** Daily stock prices from 1990, financial reporting from 2008, open-source activity from 2008, Benzinga event data from 2011, SEC filings from 2012, brand reviews from 2013. These are long enough to train on.

6.2 What Carbon Arc is not good at

These are not criticisms of the platform; they are properties of alternative data, and every professional user manages them. Your handling of them is part of what is being evaluated.

1. **Short histories.** Most alternative panels begin between 2019 and 2024. Your out-of-sample window starts in 2021. A panel that starts in 2019 therefore gives you about two years of pre-sample history to train on, which is not enough to fit anything complex without overfitting it. Check the start date before you build a feature, not after.
2. **Heterogeneous and sometimes metric-specific lag.** Publication lag across the library runs from four hours to ninety days, and within a single asset it can differ by metric — one asset publishes brand-level weekly data

at T+9 days, category-level weekly at T+16 and monthly at T+12. A blanket "lag everything one month" rule is wrong in both directions: too conservative for some series, and still leaking for others.

3. **Revisions and backfill.** The series you see today is not what you would have seen at the time. Panels are backfilled as new contributors are added and restated as data is cleaned. If you use the current vintage of a revisable series as a historical feature, you have leaked — and it will not look like a leak in your code.
4. **The entity map is a present-day map.** Mappings are built against the companies that exist now. Firms that were acquired, went private or were delisted during your sample may be missing or incompletely mapped. This is a survivorship problem and it is the one most likely to bite you (Section 6.3).
5. **Panel composition drift.** A card panel that captured 2% of a retailer's sales in 2019 may capture 6% today. The "growth" that falls out of the raw level is an artefact of the panel, not of the company. Work in shares, growth rates and cross-sectional ranks, not levels, and normalise against the panel's own total wherever you can.
6. **Sector coverage is skewed.** Consumer, retail, technology, media and healthcare are richly covered. Financials, utilities, energy and much of industrials are not. A signal that only fires in consumer names is a sector bet wearing a stock-selection costume — and in a market-neutral book, an unintended sector bet is exactly the exposure the mandate is supposed to eliminate.
7. **Some series are closed.** Several assets in the library are historical only and receive no further updates; a few end during 2025 or 2026. They are usable for research and useless for anything that has to run forward. The workbook flags them.
8. **Crowding.** These datasets are commercially licensed and widely used. A meaningful part of the obvious signal in them is already in prices. Treat every alternative dataset the way you would treat a new factor: assume it is contaminated until you have shown that it is not.

6.3 The survivorship test we expect you to run

Survivorship exposure in this library comes in three forms, and they need different treatment:

- **Immune.** Assets keyed to something other than a company and resolved to tickers only at the point of use — weather, macro series, commodity prices, box office. The underlying records do not depend on which firms still exist.
- **Crosswalk-only.** The observations themselves are complete, but the company mapping is current-vintage. Card transactions at a merchant that no longer exists were still recorded; the question is whether that merchant still resolves to a ticker. This is the large middle category, and it is fixable if you check.
- **Universe-level.** The asset only contains entities that are in some present-day index, list or panel. Here the missing names are genuinely absent, not merely unmapped, and no amount of care at the join recovers them.

A concrete test, which takes an hour and which we will ask about: take a list of U.S. companies that delisted between 2017 and 2020 — Sears, Bon-Ton Stores, Pier 1, GNC, Ascena, J.C. Penney are a reasonable starting set — and count how many of them resolve in the entity map of each asset you intend to use. Then run the complementary return test: within your universe, compare the average return of names your dataset covers against names it does not, over the same period. If covered names systematically outperform, your coverage is not random and any signal built on it inherits the bias. Report what you found. A team that documents a coverage bias and controls for it will score better than a team that did not look.

6.4 The 8-K corpus: a worked example from the first round

The previous section describes a test. Here is what happened when one team actually ran it, on a dataset we supplied ourselves.

In the first challenge we provided a corpus of 8-K filings covering the sample period. Out of multiple submissions, only one team checked whether that corpus covered companies that had left the universe during the sample — and found that it did not. Firms that delisted, were acquired or went private between 2015 and 2026 were materially under-represented, because the corpus had been assembled against a present-day list of filers rather than against the filers who existed at each point in time.

Three things are worth saying plainly about that.

It was a real defect, and the finding was correct. The corpus we handed you carried a survivorship bias. A text signal estimated on surviving firms only will overstate its own performance, and it will do so in a particular direction: the companies that disappeared are disproportionately the ones whose news was bad. A model trained to read 8-Ks learns, without anyone intending it, that disclosures of this kind are usually survivable — because in the data it was given, they always were.

Most of teams did not ask. That is the part to sit with. The corpus arrived from a credible source, with a tidy schema, in a competition where the organisers had clearly put work into the data. None of that is evidence of completeness, and the question costs an hour to answer. This is precisely the test in Section 6.3, and the reason it is in this document is that almost nobody ran it.

There is a clean remedy, and you have it this year. See below.

EDGAR is survivorship-free

The SEC's EDGAR system is an archive of what was filed, when, and by whom, keyed to the filer's CIK. Nothing is removed from it when a company delists, is acquired, or goes bankrupt. The filings of a company that ceased to exist in 2018 are still there, exactly as filed, and will remain there permanently. A text channel built directly from EDGAR therefore inherits no survivorship bias from the text side — and this is true of 10-K and 10-Q filings just as it is of 8-Ks.

Carbon Arc's **CA0032, Financial Filing Forms**, is EDGAR: roughly 150,000 filings a year across ten form types, from 2012 onward, keyed by CIK. It gives you the option of rebuilding the text channel from the primary source rather than from a curated extract — on a footing you can verify yourself rather than take on trust.

One caveat, so that you do not over-claim this in your deck:

- **Map through CIK, not through ticker.** CIKs are stable and permanent. Tickers are neither — they change, and they get reassigned to unrelated companies after a delisting. A ticker-based join over a long sample will silently attach one company's filings to another company's returns.

Which filings to use

The two form families do different jobs, and the best submissions will probably use both. A 10-K or 10-Q is periodic, long, comprehensive and heavily lagged — the right source for fundamentals, management discussion, risk-factor language and year-over-year changes in how a company describes itself. An 8-K is event-driven, short, tied to one occurrence and timestamped to the day, which makes it far better suited to the daily decision frequency this year's challenge permits. Use the 8-K for timing and the periodic filings for content, and be explicit in your deck about which filings fed which part of your signal.

6.5 The news data — read this before you plan around it

You asked, in effect, whether you can form signals from daily news as it arrives. The honest answer requires

knowing what the news asset actually contains, and it is not what most teams will assume. The relevant asset is **CA0035, "Financial News & Data", provided by Benzinga**. Its properties, verified against the catalogue:

Property	Value
History	2011 – present
Coverage	3,000+ tickers (Wilshire 5000 plus roughly 1,000 widely-held names outside it)
Update frequency	Weekly
Publication lag	T + 1 day
Record granularity	Date (calendar day). No intraday timestamp.
Fields	Ticker · event label · exclusive-vs-non-exclusive mention flag · date
Event taxonomy	Earnings Beats, Price Target, Offerings, Acquisitions, Contracts, SEC, Movers, Pre-market Outlook, Hedge Funds, Federal Reserve, Guidance, Politics, Sector ETFs, General, and others
Also included	Analyst rating, earnings, dividend, IPO and M&A calendars; bull/bear viewpoints
Not included	Headline text · article body · intraday timestamp · sentiment score

Three consequences, and please plan around them rather than discovering them on day one.

1. There is no news text here. This asset is an event-tagged table, not a corpus. You cannot run an LLM over headlines or article bodies, because neither is in the data. If your plan involves language models reading news, your text source is the 8-K corpus from the first challenge, or the SEC filings asset (CA0032) — both of which do contain full text. What this asset gives you instead is clean, long-history, ticker-keyed event counts by type, which is a different and in some ways more tractable signal.

2. Daily signals yes, live daily feed no. Records are dated to the calendar day, so you can build daily event-based features across the full history for backtesting. But the platform refresh is weekly, so during the two-day event you will not be receiving a live daily news stream. Design for a historical daily backtest, not for live trading on today's news.

3. No intraday timestamp is a leakage trap. An event dated day t may have been published before the open, during the session, or after the close — the data does not say which. You must therefore treat any event dated t as actionable no earlier than the open of day t+1. Trading at the close of day t on an event dated day t is a look-ahead violation, and **it is the single most likely way a team may disqualify itself this year**.

One genuinely useful field deserves a mention: the **exclusive-mention flag** tells you whether the underlying article was about this ticker alone or mentioned it among many. That is a reasonable proxy for whether the event was material to the company or incidental, and it is the kind of field that separates a thoughtful feature from a naive event count.

Two clocks, and which one governs. The T+1 figure above is the source publication lag; the weekly figure is how often the platform delivers. For this competition, historical eligibility follows source publication: an event dated day t may enter a feature from day t+1, under the execution convention in Section 7. We allow this because the backtest is the exercise, and the weekly cadence is an artefact of delivery rather than of when the information

existed. A live implementation, though, would see the week's events only at the next refresh — so if your signal's edge decays inside a week, say so, and show what weekly-delivery timing would do to it. A strategy whose alpha survives only under source-publication timing is a strategy that could not have been run.

6.6 Suggested starting points

From the full catalogue, these are the assets with a usable publication lag, direct or near-direct ticker keying, and enough history to be worth the effort. Most update daily; the exceptions are marked as such in the table — CA0035 and CA0043A are weekly, CA0015 monthly, CA0065C quarterly, and CA0032 is an event feed — and they earn their place on history and content rather than on frequency. The workbook has the complete list with verdicts; this is where we would start. One asset breaks the training-history rule deliberately. CA0086 begins only in 2023, well inside the out-of-sample window, and it is listed because it is the only intraday series in the library. A series that starts after the beginning of your test period may still be used, on two conditions: it enters the strategy only from the date its history begins, with the walk-forward respecting that start rather than back-filling it, and you report the sub-period in which it was active separately from the full-period result. Do not fit a model on a 2023-onward series and then present a 2021-onward backtest of it.

ID	Asset	Provider	History	Freq.	Lag
CA0092	Securities Ratings	New Constructs	2016–	Daily	T+2d
CA0065S	Stock Prices	Fiscal AI	1990–	Daily	T+1d
CA0035	Financial News & Data	Benzinga	2011–	Weekly	T+1d
CA0032	Financial Filing Forms	SEC EDGAR	2012–	Feed	T+1d
CA0056	Credit Card – US Complete	Facteus	2019–	Daily	T+3d
CA0058	Credit Card – Health Spend	—	2017–	Daily	T+5d
CA0030	Clickstream	—	2020–	Daily	T+1d
CA0060	Foot Traffic	PassBy	2019–	Daily	T+5d
CA0029	POS – Convenience Stores	PDI Technologies	2019–	Daily	T+1d
CA0052	Brand Reviews	—	2013–	Daily	T+1d
CA0098	Open-Source Dev Activity	Sherlock Data	2008–	Daily	T+1d
CA0065C	Company Reported Actuals	Fiscal AI	2015–	Qtrly	T+1d*
CA0043A	Job Openings	PredictLeads	2017–	Weekly	Bi-mo.
CA0086	Credit Card – US Neo	—	2023–	Intraday	T+4h
CA0015	Macroeconomic Data (FRED)	St. Louis Fed	1980–	Monthly	~real-time

** After the company releases. CA0092 updates Sunday through Friday. CA0086 is the only intraday asset but has the shortest history.*

Before you commit to any asset, answer four questions in writing: (i) when does the history start, and is that

early enough to train before 2021? (ii) what is the publication lag for the specific metric I am using? (iii) is this series revised, and if so can I reconstruct the vintage? (iv) what fraction of my universe does it actually cover, and is that coverage random with respect to returns? If you cannot answer all four, do not use the asset. Three well-understood datasets beat eight you are guessing about.

7. Forward-Looking Bias

Critical rule. If your portfolio construction, signal generation, model selection or optimisation uses forward-looking information — intentionally or accidentally — your team will be disqualified. At any point in time t you may use only information that would genuinely have been available at t.

In most of data science, using all available data is acceptable. In finance it is not, because the whole claim you are making is that your method would have worked at the time. How we verify that claim has changed this year — we examine the economics of your portfolio rather than audit your code — and Section 11 explains what that means in practice. The short version: a leak shows up in the behaviour of the returns, and that is much harder to conceal than source code.

The examples below are the ones we actually find. The first two carry over from previous editions. The rest are specific to daily decisions and to alternative data, and they are where we expect this year's problems to be.

The classic two

1. **Feature selection on the full sample.** Running importance analysis over 147 characteristics across the whole 2015–2026 panel, identifying the factors that mattered, and then training a model from 2015 using only those factors. By 2018 you could not have known which factors would matter through 2026. You have used future knowledge to improve past decisions.
2. **Regime identification with future data.** Defining high- and low-volatility regimes from the full VIX series and then applying those labels retroactively. A regime-switching model must classify each date using only data up to that date.

The daily-frequency and alternative-data cases

1. **Same-day news without an intraday timestamp.** Covered in Section 6.5 and repeated here because it is the most likely violation this year: an event dated day t is actionable no earlier than the open of day t+1. Because the supplied daily returns are close-to-close, you cannot simulate an opening trade exactly. Use this convention, and state any departure from it: information dated through day t forms the book at the close of day t+1, and the first return you earn on it is the close of t+1 to the close of t+2. The same convention applies to any feature computed from day-t information, including the day-t return itself.
2. **Earning a return your signal already used.** If a feature uses the day-t return — a reversal signal, or a momentum or volatility measure computed through day t — then that day's return is already spent, and you cannot also earn it. Teams routinely compute a signal through day t and then book the day-t return as strategy performance, which silently credits them a day they could not have captured. Under the convention above, a book formed on information through day t first earns the return from the close of t+1 to the close of t+2.
3. **Current-vintage alternative data.** Pulling a revisable panel as it stands today and using it as a 2021 feature. Lagging the date does not fix this. A lag handles publication delay; it does nothing about a value that was revised upward six months later, because the revised number is what sits in the cell whatever date you attach to it. Only three treatments are acceptable: reconstruct the vintage from the asset's versioning columns; or show that revisions to the series cease within a bounded interval and lag beyond it, with the evidence in the

deck; or leave the series out of the point-in-time backtest and use it only descriptively. If you cannot do one of the three, do not build a feature on it.

4. **Survivorship through the entity map.** Building a feature that exists only for companies the vendor can still map, and comparing it against a universe that includes companies it cannot. Section 6.3.
5. **Event date versus announcement date.** Calendar-type data is especially dangerous here. A scheduled earnings date is known in advance and is legitimate. An analyst rating or a price target is known only when published. Check which date a field carries before you align it to a formation date.
6. **Delisting and the missing stub return.** At daily frequency this matters more than at monthly. If a name simply disappears from your return file on the day it collapses, a short position in it books the gain on the way down and never takes the final stub. State explicitly how you handled delistings, halts and missing days.
7. **Full-sample standardisation.** Z-scoring a feature using the full-sample mean and standard deviation, or winsorising at full-sample percentiles. Normalise cross-sectionally within each date, or with an expanding window — never with statistics computed over the whole period.
8. **Hyperparameters tuned on the test period.** Choosing a learning rate, a penalty, a lookback or a number of holdings by what performed best out of sample. This is the most common form of leakage in the whole competition and it almost never appears in the code as a date error. Tune on validation data that precedes the test window, and keep the split dates explicit.

Model memory: the leak you cannot instruct away

One finding from the first round deserves its own treatment, because it breaks an assumption almost everyone starts with.

Several teams discovered that a language model instructed not to use forward-looking information still used it. Told explicitly to reason only from the text in front of it, the model nonetheless recognised the company from that text and brought to bear what it already knew about how the story ended. One team went further and blinded the filings — stripping the company name, the ticker and other obvious identifiers — and found that the model frequently re-identified the firm anyway, from the residue that a document inevitably carries: the business described, the products named, the geography, the dollar magnitudes, the particular phrasing of a known event.

Take the implication seriously. A pretrained model is not a blank reasoner that you point at your data. It is a system that has already read a great deal about the companies in your universe, including what happened to them after the date you are pretending it is. Two things follow that are easy to get wrong:

- **An instruction is not a control.** "Do not use information after this date" in a prompt is a request, not a mechanism. The model has no reliable way to partition its own knowledge by date, and no way to tell you when it failed to.
- **Blinding identifiers is not sufficient either.** It raises the cost of re-identification; it does not eliminate it. A filing stripped of its ticker is still a highly specific document.

This is an active research question rather than a settled one, and the evidence cuts both ways — smaller models appear markedly less prone to it than large ones. We are not banning pretrained models; the competition would be pointless without them. What we are asking is that you stop treating the problem as solved by a line in a prompt.

What we expect you to do about it

1. **Report your model's release date against your test window.** If the model was released partway through the out-of-sample period, split the results: performance before the release date and after it. The post-release sub-period is the cleaner test, and a strategy that holds up across the boundary is making a much stronger claim than one that does not.

2. **Run the re-identification test, and report the number.** Take a sample of your blinded inputs, ask the model to name the company, and measure how often it can. That hit rate is a direct estimate of how much of your "blinding" is real. It takes an hour, nobody has a good answer for it, and a team that reports it honestly — whatever the number — will be taken more seriously than a team that asserts the inputs were blinded and moves on.
3. **Prefer smaller or older models where the task allows.** If a compact model does the classification or extraction adequately, it carries less of this risk than a frontier model, and you can say so with evidence rather than hope.
4. **Build at least one signal that does not depend on a pretrained model at all.** A structured feature from the panel or from Carbon Arc has no memory of 2023. Having one in the book gives you a clean comparison and a fallback if the text channel is challenged.

How we will treat this. Residual model memory is a limitation of the current technology, not misconduct, and we will not disqualify a team for using a pretrained model. We will treat an undisclosed one quite differently from a measured and disclosed one. If your deck claims the text channel is free of look-ahead because the model was instructed not to look ahead, the first-round evidence says that claim is wrong, and the judges now know it.

Cost and borrow assumptions: a disclosure rule, not a leak

One case from the first round deserves its own rule, because it is genuinely ambiguous and we would rather settle it in advance than argue about it afterwards.

Modelling trading costs or borrow fees requires a fee schedule, and published schedules are current — nobody has a 2021 vintage of a broker's commission table or short-stock file. Calibrating a cost model on today's schedule and applying it across your sample is therefore permitted, and it is better practice than assuming shorting is free. But it is a forward-dated parameter choice, and it must be handled openly:

- State in the deck which schedule you used, as of what date, and what the model does not capture (market impact, locate failures, recalls are the usual omissions).
- Report your headline result with a flat, conservative alternative assumption alongside it — for example a uniform borrow fee of 50 bp for large caps and 5% for the rest. If the strategy only works under your fitted curve, that is important and we need to see it.
- Never let a cost or fee parameter enter the return forecast. It belongs in the construction and ranking layer only.

Disclosed modelling assumptions are judged on merit. Undisclosed ones are judged as leakage. Nothing in this section is intended to discourage you from modelling implementation costs — the teams that did so in the first round produced visibly more realistic portfolios. Model them, and write down what you assumed.

8. Methodology Standards

Every team in this field is working with AI coding assistants, and several of you are building agentic research workflows as the strategy itself. That is the point of the competition and we are not going to constrain it. But it raises the bar on process discipline rather than lowering it. A model that cannot be reproduced is not a result, and an agent that quietly read the wrong file has leaked whether or not anyone wrote a line of leaking code.

The standards below are what a quantitative research team would expect of its own members.

8.1 Walk-forward discipline

1. **State your split dates explicitly,** in the deck and in the code. The first-challenge convention is a reasonable default: train 01/2015–12/2018, validate 01/2019–12/2020, first out-of-sample year 2021, then expand the training window and roll the validation window forward one year at a time.
2. **Refit on a schedule, not on a hunch.** Annual or quarterly refitting with an expanding window is standard. Refitting whenever performance dipped is curve-fitting.
3. **Write a leakage test.** At minimum, one assertion per data join that the maximum date on the feature side is less than or equal to the formation date, accounting for publication lag. This is five lines per join and it catches the majority of real leaks. We will not be inspecting these tests — **write them because they are how you find a leak before the judges find its signature in your returns.**
4. **Separate the modules.** Forecast, construction, and cost accounting should be distinct components with a defined interface. Beyond good practice, this is what makes it possible for you — and for us — to see which layer a result actually comes from.

8.2 Working with agents and language models

1. **The agent sees only what the strategy sees.** If an agent reads a document, that document must be dated on or before the formation date. If it searches the web, it will find the future — so either disable retrieval or constrain it to a dated corpus, and say which you did.
2. **Log every prompt with the formation date it belonged to.** The prompt log is your evidence of point-in-time integrity.
3. **Declare your model's training cutoff.** A pretrained model may have seen your out-of-sample period during training, which is a subtle but real form of look-ahead. If your model was released partway through your test window, report the sub-period after its release separately. A team in the first round did this well and it materially strengthened their submission.
4. **You own what the agent wrote.** Read the code it produces. "The assistant generated it" is not a defence against a leak, a wrong sign, or a fabricated number — and agent-written code leaks in characteristic ways, most often by reaching for a convenient full-sample statistic.

8.3 Honesty in the numbers

1. **Every figure in the deck traces to an output file.** No number typed in by hand. If a judge asks where 1.42 came from, you should be able to name the file.
2. **Report what broke.** If you breached a constraint, exceeded your turnover budget, or found that a signal stopped working in 2024, say so, with the economic reason and what you would do with more time. This reads as competence, not weakness.
3. **Name what you did not model.** Market impact, borrow availability, short recalls, financing spreads, capacity. You are not required to model all of them. You are required to know which ones you skipped.

9. Standardised Performance Reporting

Last year teams reported performance in a dozen different formats, which made comparison across submissions harder than it should have been. This year the reporting format is fixed. The table below is what an allocator expects to receive, and every team will submit it on one slide, for the full out-of-sample period, for the strategy and for the benchmark where applicable.

9.1 The required table

Metric	Units / basis	Strategy	Benchmark
Ending value of the endowment	US$ billion		
Cumulative return over the period	%		
Annualised return	% per year		
Annualised volatility	% per year		
Annualised alpha vs S&P 500	% per year (+ s.e.)		n/a
Beta vs S&P 500 — monthly	coefficient (+ s.e.)		n/a
Beta vs S&P 500 — daily	coefficient (+ s.e.)		n/a
Sharpe ratio	annualised		
Information ratio vs T-bill + 4%	annualised		n/a
Tracking error	% per year		n/a
Maximum drawdown	%		
Maximum one-month loss	%		
Maximum one-day loss	%		
Positions held	average / min / max		n/a
Gross exposure	average / maximum, % of capital		n/a
Net exposure	average / min / max, % of capital		n/a
One-way turnover	% per rebalance and annualised		n/a
Average holding period	trading days		n/a
Average position size	US$ million		n/a

Metric	Units / basis	Strategy	Benchmark
Short book median market cap	US$ million		n/a
Trading cost	% of NAV per year		n/a
Borrow cost	% of NAV per year		n/a
Information ratio, net of all costs	annualised		n/a

9.2 Conventions

Annualization — and two different numbers. State the frequency you worked in. From monthly data, multiply the mean by 12 and the standard deviation by √12; from daily data, by 252 and √252. Do not mix the two within one statistic. These are arithmetic annualised moments, and they are the inputs to the Sharpe and information ratios. They are not the compound growth rate, and they will not reconcile with the ending value of the endowment. Report both, and label them: the CAGR — (ending NAV / starting NAV) raised to the power 1/years, minus one — is the figure that ties to the dollar outcome; the arithmetic mean is the figure that drives the risk-adjusted ratios. A strategy with meaningful volatility will show a CAGR visibly below its arithmetic annualised return, and that gap is not an error.

Active return and information ratio. Define the active return in period t as your portfolio return less the benchmark return for that period, where the monthly benchmark return is b_t = TB3MS_t / 1200 + 0.04 / 12, with TB3MS_t the annualised 3-month bill rate in per cent for that month, and the daily equivalent is b_t = TB3MS_t / 25200 + 0.04 / 252. A benchmark series computed on this basis will be supplied with the data, so that every team is measured against identical numbers. The information ratio is the annualised mean active return divided by its annualised standard deviation. This is the headline number for the mandate.

Alpha and beta. Regress your excess return on the S&P 500 excess return. The intercept, annualised by multiplying by the number of periods per year, is alpha; the slope is beta. Report the standard error (or t-stats) on both. For a market-neutral book the beta is evidence of compliance with the mandate and the alpha should account for very nearly all of your mean excess return. Report beta at both daily and monthly frequency: a book that looks neutral monthly can carry meaningful daily beta, and vice versa.

9.3 Turnover — the definition we will use

Turnover was reported inconsistently last year, so here is the formula. On each rebalancing date τ, one-way turnover is

$$\text{Turnover}(\tau) = \tfrac{1}{2} \cdot \Sigma_i \left| w_{i,\tau} - w_{i,\tau^-} \right|$$

where $w_{i,\tau}$ is the weight of name i immediately after the rebalance and w_{i,τ^-} is the weight carried into it. The one-half makes it one-way: a position fully replaced by another counts once, not twice. Report the average per-rebalance figure and the annualised figure, which is the average multiplied by the number of rebalances per year — 252 for daily, 52 for weekly, 12 for monthly. Also report the average holding period. It is not simply the inverse of turnover for a leveraged book: at 200% gross, replacing the whole book once a year produces 200% annualised one-way turnover, so the inverse would wrongly imply a six-month holding period. Measure holding periods directly — the position-weighted average number of trading days a name stays in the book — or report average gross exposure divided by annualised turnover and label it an approximation.

For context, and noting that these figures are stated as a fraction of NAV rather than of gross exposure: for a long-only book run at 100% gross, 100% annualised turnover means the book is replaced once a year. Under this mandate, at 200% gross, replacing the whole book once a year is 200% turnover. Long-only institutional managers typically run 20–80%. Active quantitative equity strategies commonly run 100–400%. Daily rebalancing of a sizeable fraction of the book takes you into the thousands of percent, where cost assumptions stop being a detail and become the result.

9.4 Required exhibits

- Cumulative active return of the strategy against the benchmark, over the full out-of-sample period, with the S&P 500 shown for context.
- Rolling beta to the S&P 500 — a 252-day window — across the out-of-sample period.
- Calendar-year returns, before and after costs, against the benchmark.
- Time series of gross exposure, net exposure and position count.
- Top ten and bottom ten holdings by average weight over the out-of-sample period, with those average weights.
- The ten largest positive and ten largest negative contributors to return, by name.
- Short-book characteristics: distribution of market capitalisation and of dollar volume, demonstrating compliance with the US$500m floor.
- Distribution of position size in days of average daily volume, long and short legs shown separately.

10. Final Presentation

Each team has ten minutes before the panel of judges. The judges are senior asset management professionals, several of whom oversee multibillion-dollar portfolios and many of whom are deploying AI in their own investment processes. Some are not AI specialists. All of them evaluate investment products for a living.

They will be most interested in your investment thesis, your process, and the future potential of the product — not in the architecture of your model. Technical depth matters only where it connects to a financial reason. "We used a transformer" is not an investment argument; "we believed the market underreacts to supplier-relationship changes, and here is how we measured them" is.

Minimum deck contents

1. **Executive summary.** The strategy in four sentences, the methods used, and headline performance against the benchmark.
2. **Investment thesis and signals.** What inefficiency you are harvesting and why it should exist. Which signals express it, and why you chose them.
3. **Data and methodology.** Datasets used, including any Carbon Arc assets and your answers to the four screening questions in Section 6.6. Your model, your training and validation design, your split dates.
4. **Portfolio construction.** How forecasts become positions. Which definition of neutrality you enforce, how you size, how you handle the short-side market-cap floor, your rebalancing frequency and why.
5. **Risk management, and your agent evidence.** How risk is controlled at the position and book level. If you used an agent anywhere in construction or risk, show the counterfactual required by Section 5.3 — the same strategy with the agent and with a fixed rule, side by side — along with the decision log and one worked example of the agent earning or failing to earn its place.
6. **The standardised performance table** from Section 9.1, plus the required exhibits from Section 9.4.
7. **Costs and implementation.** Your cost assumptions, performance before and after costs, and the sensitivity

check described in Section 7.

8. **Discussion.** Did the strategy behave as you trained it to? Which signals drove performance, which positions, and which macroeconomic events? Where did it fail, and what would you do with more time? Any constraint breaches, with their economic rationale.

11. Data Access

Data

Core panel. You continue with the dataset from the first challenge: the 147 characteristics and the 8-K text corpus.

Daily returns. A new file, dsf_mini_from_2014.csv, containing daily returns for the universe from 2014 onward. Merge it to the monthly panel on permno and control for date, do not merge on forward date. Once a name delists it simply has no further daily return data. If your strategy genuinely needs daily price levels — for a liquidity or participation measure, say — they are available through Carbon Arc (CA0065S, daily since 1990) and through Arcana.

A note on the 8-K corpus. The supplied 8-K file carries the survivorship defect described in Section 6.4. It stays in the folder because many of you have already built against it, but if you use it, acknowledge the limitation in your deck and say how you handled it. Rebuilding the text channel from EDGAR (Carbon Arc CA0032) avoids the problem entirely and will be credited.

Carbon Arc. Onboarding instructions and credentials will be sent separately.

Screening workbook. *MCGILL_2026_CarbonArc_Data_Screen.xlsx* — the catalogue manual described in Section 6.

How your work will be reviewed: the Arcana platform

Every team will be given an account on **Arcana** (www.arcana.io), the second platform you will work on this year. Arcana serves three purposes at once: it is a tool you use while building, it is the lens through which the judges will examine your portfolio, and it is where you will present. Full instructions will come during the two-day event; what follows is enough to understand what is being measured, and to think about it before you arrive.

One account per team. Unlike Carbon Arc, where every team member is onboarded individually, Arcana issues a single account per team, registered to the team leader's email address. Team leaders are responsible for giving their members access to it. Onboarding emails from both platforms arrive over the coming days — Carbon Arc first, Arcana at the same time or shortly after. Do not let them sit unread in a spam folder.

Arcana is a portfolio intelligence platform used by hedge funds, multi-manager platforms and allocators. The simplest way to place it: if you have seen factor analysis run through Bloomberg, or a Barra risk report, Arcana does that and considerably more. It carries a library of over 120 risk models — MSCI Barra, Axioma and Wolfe Research among them, alongside its own proprietary factor models — with roughly twenty years of model history, so it covers the whole of your out-of-sample period. Where a conventional risk report gives you a static snapshot of exposures, Arcana decomposes performance continuously: factor contribution against idiosyncratic contribution, position by position and period by period, with the attribution recomputed as the portfolio evolves.

The reason we are using it is that it answers the questions this mandate actually turns on, and a factor report does not.

And it connects to Claude. In October 2026 Arcana released an MCP connector that puts portfolio, risk,

performance and research data directly into Claude, so that a model can be asked about a book in natural language and answer from the portfolio itself rather than from generic knowledge. Arcana's own examples include summarising the major sources of risk across a portfolio, identifying changes in exposures, explaining recent performance, understanding thematics and crowding, comparing positioning against a benchmark, and exploring hedge ideas. The announcement, 6 October 2026, is here: Arcana Brings Institutional Portfolio Intelligence Directly to Claude.

Take that seriously when you plan your two days and the current time you have before October 28 and 29, because it turns what Section 5 asks of you from an ambition into something buildable. The hardest part of putting an agent into the risk layer is normally giving it a trustworthy view of the book — exposures, attribution, factor decomposition — and that is exactly what this connector supplies, computed by an institutional risk engine rather than by code you wrote at two in the morning. An agent that reads your live exposures through Arcana and reasons about what to do next is the shortest path we know to the kind of submission described in Section 5.1.

What the judges will be looking at

1. **Factor versus idiosyncratic return.** How much of your performance is exposure to known factors, and how much is genuinely idiosyncratic? For a market-neutral mandate this is close to the whole question. If your information ratio is being produced by a persistent small-cap tilt or a value tilt, you have not built an alpha product — you have built a cheap factor exposure with an expensive process attached, and an allocator can buy that elsewhere for a few basis points.
2. **Unintended exposures, and when they emerged.** Arcana will show not just that your book drifted into an exposure but the date it started and what drove it. "We were sector neutral" is a claim the platform will either confirm or contradict in about ten seconds.
3. **Sizing versus selection.** Did you make money by picking the right names, or by sizing them well? These are different skills, they come from different parts of your pipeline, and most teams have never separated them. If your returns come from sizing, your signal may be weaker than you think and your construction better.
4. **Hit rate against win/loss ratio.** A book that is right 48% of the time with a 1.6 win/loss ratio is a completely different animal from one that is right 58% of the time with a ratio of 1.0. Both can produce the same information ratio, they fail in different market conditions, and you should know which one you have built before a judge tells you.
5. **Alpha decay.** How quickly the edge in your signal dissipates after formation. This is the empirical answer to the rebalancing-frequency question in Section 4 — if your alpha decays over three days, a monthly book is leaving most of it on the table; if it decays over two months, daily trading is paying costs for nothing.
6. **Earnings versus ex-earnings performance.** Particularly relevant this year, given how many of you are building event-driven signals from 8-K filings and news. If essentially all of your alpha arrives in earnings windows, you have a specific and capacity-constrained strategy rather than a general one. That is not a criticism, but it has to be the strategy you say you built.
7. **Stress and scenario behaviour.** The platform carries more than fifty stress scenarios spanning factor shocks, macro events and historical episodes. Expect to be asked how the book behaves in a few of them, including ones outside your sample.

Why this catches what a code review would not

We are not going to audit your code. Reading eleven repositories in two hours, even with AI, was never a realistic way to catch anything.

The economic review is the more demanding test, not the lighter one. A data leak does not look like a bug in the economics — it looks like a strategy that works for no reason anyone can articulate. The idiosyncratic share of

return is implausibly high with no identifiable driver. The hit rate is better than any professional book achieves. The returns concentrate in exactly the periods or names where the leaked information would have been most valuable. The exposures do not match the thesis in the deck. These are signatures, they show up in the attribution, and unlike a line of code they cannot be tidied up before submission.

The practical consequence is that the discipline in Section 8 is now for your benefit rather than ours. We will not be checking whether you wrote the leakage tests. We will be checking whether your results behave like results from a strategy that does what you say it does. Write the tests so that you find the problem first.

Be ready to answer, for any position or any period a judge points at: why was this trade on, what did the strategy believe, and what did it earn? A team that can do that fluently is in a very strong position. A team that cannot is in trouble regardless of the information ratio.

Prepare for this lens now. You do not need Arcana to ask these questions of your own work — you need only decompose your returns into factor and residual components, split your P&L by sizing and selection, and compute your hit rate and win/loss ratio. Do it before the event. The teams who are surprised by what the attribution shows will be the ones who never looked.

What you will need to submit before final presentation:

- Presentation deck. Decks are presented with Arcana generated output and out further instructions; the format and upload instructions come at the event.
- Holdings file: date, identifier, weight, for every rebalancing date in the out-of-sample period. This is what drives the economic review, so the format matters — the specification will come with your Arcana instructions.

12. Closing

The coding language of the competition is Python. Comment explicitly and keep a consistent style — not because we will be grading the code, but because you will be explaining your own strategy under questioning, and a pipeline you cannot follow is one you cannot defend.

We do not expect you to reinvent portfolio management in two days. We expect rigour, honesty about what you do not know, and an investment idea that a professional would recognise as an idea. The finalists who have gone furthest in previous editions were not the ones with the most elaborate models; they were the ones who could explain precisely why their strategy made money and under what conditions it would stop.

Further constraints and the final data configuration will be issued at the start of the event. Start working now. Good luck.